

RESEARCH BRIEFING · 14 AUG 2026

Cerberus

A two-tier, kernel-level DDoS defence that stops attacks without punishing the real users caught in a traffic surge.

3

Novelties
demonstrated

86%

ML accuracy,
real CIC-IDS-2017

0 / 12

Legit. users
wrongly blocked

53pp

Thesis draft
written

Fast defences are blind. Smart defences are slow.

In-kernel filters

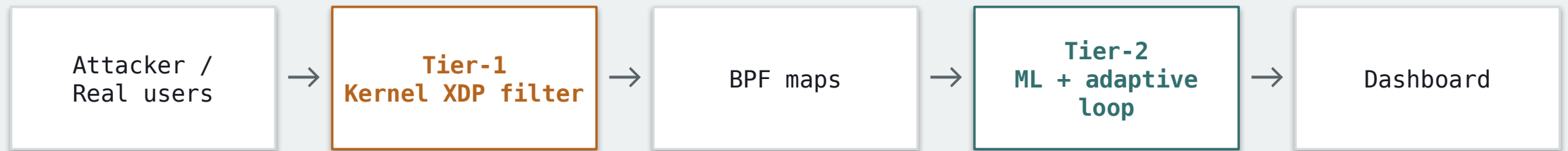
Extremely fast (run at the earliest point in Linux) but restricted to cheap, volume-only rules — cannot tell an attack from a legitimate surge.

ML-based detectors

Discriminate well, but are usually applied to ALL traffic in userspace — too slow to sit on the packet path.

The gap: no in-kernel, line-rate defence applies ML selectively enough to stay fast — AND is evaluated on the false positives it inflicts on a legitimate flash crowd.

Two tiers, joined by shared kernel memory



Every packet is counted and judged against a live threshold **inside the kernel** — the fastest point in Linux.

Only **ambiguous** sources are escalated to the ML tree. The control loop writes an updated threshold straight back into the kernel map — **no reload**.

The combination is the contribution

I

ADAPTIVE THRESHOLD

Live, reload-free updates

A control loop rewrites the kernel's drop threshold via a shared BPF map. The next packet sees the new value instantly.

II

SELECTIVE ESCALATION

ML only on the doubtful slice

The cheap kernel filter handles all traffic. Only ambiguous sources reach the decision tree.

III

FLASH-CROWD TEST

Measuring the harm, not just detection

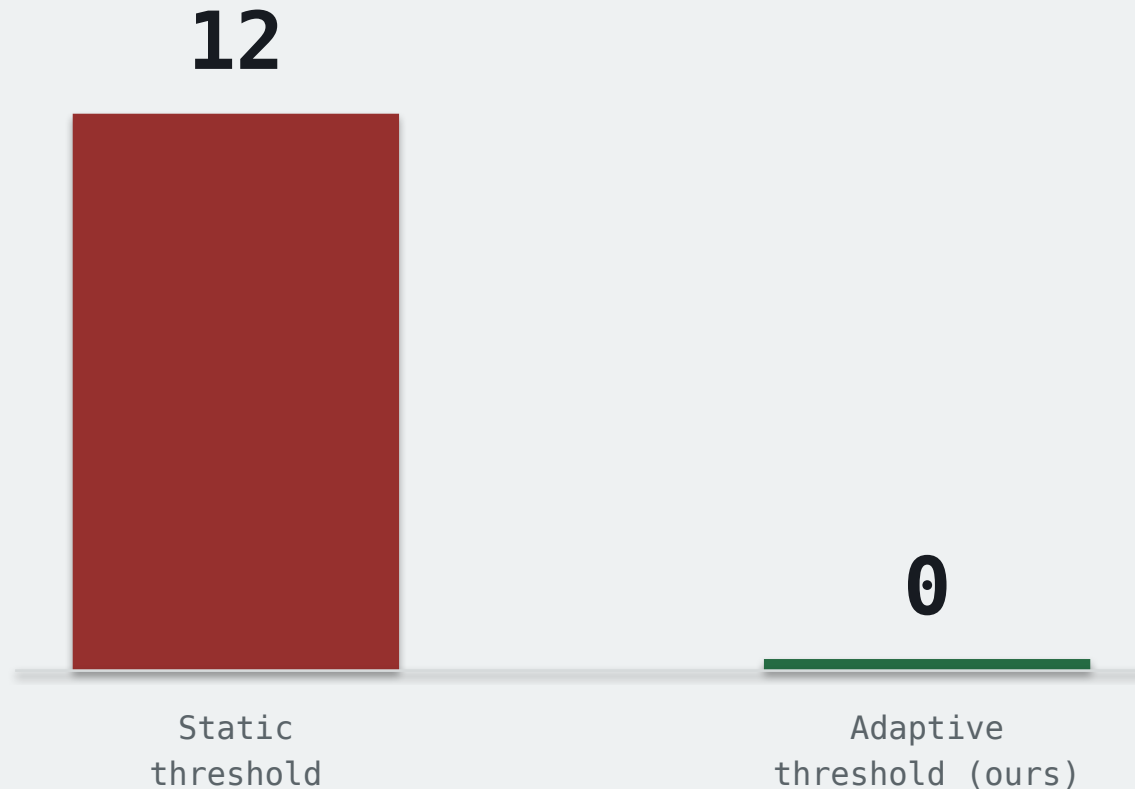
Deliberately fires legitimate surge traffic and measures false positives — what most prior work skips.

04 — KEY RESULT

0 of 12 legitimate users wrongly blocked

vs. 12 of 12 with a static threshold — both fully caught the attacker

USERS WRONGLY BLOCKED (OF 12)



Both configurations
caught the attacker.

Only the static threshold
harmed real users.

Tier-2 classifier, trained on real data

Decision tree trained on 702,718 real samples — CIC-IDS-2017 benchmark

	Precision	Recall	F1
Benign	0.97	0.78	0.87
Attack	0.76	0.97	0.85

Overall accuracy: **0.86**

Where the project stands

✓ DONE

- Kernel filter compiles & runs live on the dev machine
- Full stack demoed vs. a real hping3 flood — 1.66M packets dropped
- Tier-2 tree trained on real CIC-IDS-2017
- Flash-crowd false-positive result measured (12 → 0)
- Thesis draft written — 7 chapters, ~15,500 words

→ NEXT

- Repeat flash-crowd test on real machines, several runs
- Let the live adaptive loop choose the threshold automatically
- Fix the ML train/serve feature mismatch
- Add real literature-review citations & figures
- Compare numerically against a named prior method

What I'd like your steer on

- 1 Is the flash-crowd false-positive framing strong enough as the headline contribution for a paper?

- 2 Should the evaluation move to multiple physical machines before writing up final numbers?

- 3 Is this realistically a workshop-paper target — and if so, which venue?

- 4 Any steer on which prior papers to prioritise in the literature review?